

Spoken Protocol [00:00:00 - 00:01:07]

Good morning. Good morning, thank you. Please, please, please. Uh, good morning and welcome to Analysis 2. Uh, I have here a projection with my name, Joaquim Serra, the name of the coordinator, Enric Florit-Simon. Now, this is my introduction. Um, and then you know the times of the lectures obviously because you are here. Um, and I wanted to share before starting with the mathematical content some practical informations that are very important because we are a large group and the communication and everything needs to be done in a more or less structured way, otherwise — otherwise we collapse.

So, as I guess you did so as a general rule, uh, everything that applied to Analysis 1 applies to Analysis 2, unless otherwise stated. We will try to follow the same rules and then you know what to expect. In particular you will have this, uh, meta-for page *[points at the projector screen]* that I can see that from from up there is a bit difficult to see, right? But I will try to put it bigger here.

Projected Content: Course Website

The lecturer displays the course website for “*Analysis II: mehrere Variablen Spring 2026*”. It lists the lecturer (Joaquim Serra), coordinator (Enric Florit-Simon), lecture times (Mo 08:15-10:00 in ETA F 5, We 08:15-10:00 in HG F 7, Th 16:15-18:00 in ETA F 5), and information regarding exercise classes, communication via the Course Forum, recordings, and the final exam in August.

Spoken Protocol [00:01:07 - 00:02:55]

Um, so you know that you need to register for the exercise classes. Uh, for the communication, please try to use whenever possible the forum. We will look at the forum and and reply, and also your your peers can also reply to questions because many — so then we can use the collective knowledge, right? Um, there will be recordings of the lecture, but not until the later on, okay? Because uh is important for the lecture and for you, but mainly — I mean for you is your interest uh if you want to come or not, but for the lecture is very good that you come to the lecture, you ask questions, you stop me when I’m going too fast, these kind of things. You know, it’s very important. So it’s a — and also is where you meet with your peers and you can communicate, ask questions to your peers. It’s it’s it’s better for the learning.

Uh, we will tell you more about the exam later on, right? Today I will tell you a bit more what is important to be successful in the course, what to expect about this course, and uh the details and the rules of the exam will come later on. You don’t need to know now. The bonus, there is a zero — uh twenty-five bonus will be in questions that you will answer in some exercise classes. We will tell you in advance which ones, and uh essentially you need to be in the in the exercise class and answer it. And do not worry because uh the rule will be thought, in any case it’s just 0.25 points, uh but the rules will be thought so that if you miss one day or casuistics like this, it’s not that you get zero points, right? Okay.

Spoken Protocol [00:02:55 - 00:04:15]

Uh, lecture notes. There will be lecture notes similar to those in Analysis 1, and we will put them in this page, hopefully by the end of each week, because I I update them after

the lecture, mainly depending on your questions, depending on your feedback. And that's it. Uh, I think that the rules are mainly those and the organization. Is there some question uh about organization? So essentially as I said, um if you have doubts, uh is like in Analysis 1, unless I say something that is surprising to you as in Analysis 1. Is it okay, this? Yeah. Okay now, I will switch — I have some other things I want to project later, but now because we cannot record at the same time blackboard and projection, I will switch one moment to the blackboard, so I can mute I think, yeah. Okay. Now we need the light. *[The lecturer turns off the projector and prepares the blackboard]* Okay, I think that there is already on. Okay.

0.1 Analysis I vs. Analysis II: A Conceptual Overview

Spoken Protocol [00:04:15 - 00:05:30]

Okay now, just to start, um we will give a very quick — so quick that it's almost a cartoon — but a very quick overview of what is Analysis 1 and Analysis 2, right? It's very very quick. Analysis 1... is... sorry but for me it's very difficult to write big and then it's so — so if you don't see something, I will try to say it out loud everything I write, but if if something — I mean Analysis is a... so uh Analysis 1, calculus of one variable, right?

And then if I if we got to choose a representative theorem of Analysis 1, that a bit represents the course, uh what would you choose? So can you tell me, I know it's the first day, but let's start a bit of communication. Can you tell me at least two answers — so two people can tell me what is the representative — one representative theorem of Analysis 1? Yeah? *[Student answers]* Yeah, I agree. Some other representative theorem — he said the Fundamental Theorem of Calculus. Makes sense, right? Okay after — is there some other guess? Is — yeah, there? *[Student answers]* The Intermediate Value Theorem. Okay.

The Fundamental Theorem of Calculus in 1D

In Analysis I, the central object of study is the calculus of a single real variable.

Analysis 1: Calculus of One Variable

The most representative result is the *Fundamental Theorem of Calculus* (FTC):

$$\int_a^b f'(x) dx = f(b) - f(a) \quad (1)$$

Explanation of Steps: The lecturer initially writes $f(a) - f(b)$ on the board, but a student quickly spots the error. He corrects it to the standard $f(b) - f(a)$. He emphasizes that while the formula looks compact, understanding it required defining the derivative, the Riemann integral, and the structure of real numbers $\mathbb{R}$.

0.2 The Fundamental Theorem of Calculus in Higher Dimensions

Spoken Protocol [00:05:30 - 00:06:23]

I think we need to use this for the recording otherwise I will repeat. Later on when I get better with the throwing the *[referring to the catch-box microphone]* we will use that. So, okay, so fundamental — so we we said two theorems that are um that are indeed important. Perhaps the most representative I would agree and and I saw many people saying yes, so it's the Fundamental Theorem of Calculus, right? And it's this *[points at Equation (1)]*. Is it correct? Yeah, looks like, no?

Now, to to unfold and to write and to understand every detail about this equation over here, you needed to introduce a lot of things. You needed to define what is that (i.e., the derivative $f'(x)$), right? What is this (i.e., the integral $\int_a^b \dots dx$)? And even — and you spent a lot of time, right? What is that (i.e., the real number $\mathbb{R}$)? It was not — it was not easy. Yeah, there is some mistake? *[Corrects the board]* Good, well spotted. $f(b) - f(a)$. Now?

Okay so this is a fairly compact theorem, yet the ingredients you had to learn to unfold all of this and to understand the meaning of every symbol here were a bit of work. Now, how will Analysis 2 look like? So this will be calculus, still calculus, in uh more than one variable. Well, since you already know one, let's say at least two variables, because otherwise it's already covered by Analysis 1.

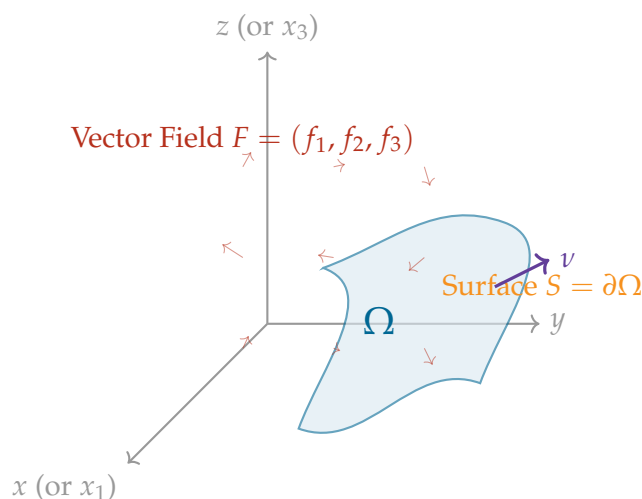
Preview: The Fundamental Theorem in Higher Dimensions

Analysis II extends these concepts to multiple variables, typically in $\mathbb{R}^n$.

Analysis 2: Calculus in ≥ 2 Variables

The higher-dimensional counterparts to the FTC are known by various names:

- Gauss's Theorem
- Stokes's Theorem
- Divergence Theorem
- Green's Theorem



The generalized theorem relates the integral of a derivative over a volume to the flux across its boundary:

$$\underbrace{\int_{\Omega} \sum_{i=1}^n \frac{\partial f_i}{\partial x_i} dx}_{\text{Integral of "Derivative" over Volume}} = \underbrace{\int_S F \cdot \nu dS}_{\text{Flux across Boundary Surface}} \quad (2)$$

Spoken Protocol [00:06:23 - 00:07:45]

And some representative theorem that would be the equivalent, so the counterpart of this one, is the Gauss — it has many names and it's always the same theorem, so but — so it's the Gauss, Stokes, uh Divergence, sometimes it's called the Divergence Theorem, uh and that's it, I think Green, sometimes Green Theorem.

And now I will write the theorem and you will not understand exactly what each thing means, I will tell you intuitively, and then the goal of the lecture will be at the end of the lecture you will be able to understand in full precision everything that I wrote here. Okay? So the theorem goes as follows. So we have — we have the space, the space, let's say I will put it in the space $\mathbb{R}^3$. Three coordinates of space, right? This thing: x, y, z or x_1, x_2, x_3 . Sometimes x_1, x_2, x_3 . Right?

And then in the — in this space we are given a domain, we are given some or some surface that encloses some interior, so this — this is the interior of the surface and the exterior. And this is a surface. Right? The surface I call it S . What is a surface? It could be a sphere, it could be a sort of the “*skin of a potato*”. It's a surface. Two-dimensional object. It has and — and what I'm — this kind of surface will have a interior and exterior, and the interior of the surface, so the — if the surface is the skin of the potato, the — the filling of the potato, the white part, right? is Ω , I call it Ω .

Insight: The Potato Analogy

The lecturer uses the “*potato*” analogy to ground abstract topological concepts. The “*skin*” represents the 2D boundary surface S , while the “*filling*” represents the 3D interior volume Ω . This physical metaphor helps students visualize the relationship between a set and its boundary in higher dimensions.

Spoken Protocol [00:07:45 - 00:09:01]

And then the equivalent of this theorem in this setup is this: Given a vector field — now you don't know what it is a vector field, let me tell you, $F = (f_1, f_2, f_3)$. These are three — three functions of space, right? At each point of space you have three numbers. Here you have three numbers, here you have three numbers, yeah. This is a vector field F . And how we do we draw vector fields? Uh, we draw it like this, right? It's like a — a field of some cereal, right? but in every point you have an arrow instead of a — a plant. Okay, so this is the vector field F . And the theorem says that the integral in Ω of the derivative of the field, the i -component with respect to the x_i variable, and you sum over i , this is equal to the flux of the field across the boundary, across the S . And it's called like this: $\int_S F \cdot \nu dS$. This is the same theorem you have got here in one dimension, in several variables.

Spoken Protocol [00:09:01 - 00:10:45]

Now, the small problem is that maybe some of you in physics or — or in previous courses have seen some of this notation, probably not, I don't expect you to — we — and we will need to introduce each of these notations. These partial derivatives, what is a domain, what is a surface? What is this surface mathematically? How do you model a surface? How do you compute an integral over a surface? You know how to compute an integral over the real line, right? over a segment, but over a surface? This will need some definition and some work. And these objects that are the partial derivatives.

So here in — in one equation you have a bit the goal and the syllabus of the course. How will — how will we proceed? So first we will start looking at the space. In Analysis 1 you did everything in $\mathbb{R}$. Now we will need to introduce $\mathbb{R}^3$, but we are mathematicians so we will — well, some of us. Uh, some of you are physicist or are proto-physicist, would like to become physicist. So we will use this because we don't want to spend time proving the same theorem in — in the plane $\mathbb{R}^2$, then in $\mathbb{R}^3$ in the space, and then when you go to do general relativity you need $\mathbb{R}^4$, right? We do everything in $\mathbb{R}^n$. n is the dimension.

Spoken Protocol [00:10:45 - 00:11:00]

So what is $\mathbb{R}^n$? Is the set of n -tuples of real numbers. This is the definition, okay? And what is this? This is the same as the set of points we will denote a point x , but now x will carry n coordinates: $x_1, x_2, \dots, x_n$. Now this is the point. And these x_i are real numbers. This you know probably because you are doing well or you have done linear algebra, right? So this should be quite familiar. But still it's good that we recall. Okay.

Spoken Protocol [00:11:00 - 00:12:30]

Now, the first part of the course we will be studying $\mathbb{R}^n$ and the properties of $\mathbb{R}^n$. The same a bit as in $\mathbb{R}$ you studied the — the axiom of the supremum, right? Or the — the fact that you can take uh limits of Cauchy sequences, these kind of facts, right? The same we will need to study in $\mathbb{R}^n$. Then there will be some new things or — or things that you could also study in $\mathbb{R}$ but become very trivial and now will be much more interesting, is that the topology of $\mathbb{R}^n$.

So first, the $\mathbb{R}^n$ as a space, as Euclidean space. So this is a bit the syllabus. Two, the topology of $\mathbb{R}^n$. You will see what does this mean in detail, the topology. Then three, this object over here [*points at the blackboard*]: partial derivatives. So when you have — when you have several variables, you can differentiate a quantity with respect to each of them, right? And differential and so — the differentiation of functions of several variables. Then integration. Integration in $\mathbb{R}^n$, here, in a domain of $\mathbb{R}^n$, and on surfaces. But because we want to learn also how to integrate on surfaces, we will do a bit of introduction to differential geometry. Introduction to surfaces and “*submanifolds*”. Now, I know it’s a lot of words. But we will introduce one by one. So surface is a usual two-dimensional object, right? Typically smooth. When it is a three-dimensional object in $\mathbb{R}^4$, for instance, we call this a submanifold. You can also call it surface but — but it’s more precise the word submanifold. Or — or or some two-dimensional object in a five-dimensional space. Any combination of dimensions. And then when you know all of this, you will need to know how to integrate. Integration over $\mathbb{R}^n$ and surfaces.

The Analysis II Syllabus

The course is structured into five primary logical blocks:

1. $\mathbb{R}^n$ as Euclidean Space
2. Topology of $\mathbb{R}^n$
3. Partial Derivatives and Differentiation
4. Introduction to Surfaces and “*Submanifolds*”
5. Integration over $\mathbb{R}^n$ and Surfaces

Explanation of Steps: The ultimate goal of the course is to prove the generalized Fundamental Theorem of Calculus (Equation 2). To reach this, we must first understand the geometry of the space ($\mathbb{R}^n$), the behavior of functions within it (Topology and Differentiation), and finally how to measure volumes and “*skins*” (Integration).

0.3 The Role of Proofs in Analysis

Spoken Protocol [00:12:30 - 00:13:50]

And hopefully then you will know all of these elements. This element — this is the normal vector, I can tell you now what is this. This is just the — the vector, the unit vector, the vector of length one that is normal to the surface, right? If you have a surface, this ν is just the normal to the surface. So this is easy... um, but still we will — we will

define this in — in detail. And after we know all of this, we will be able to prove that equation over here and work with it. And see why it is useful, right? For many things. Okay.

So this is a rough plan of the course. Now I want to — I wanted to do a very small survey about your attitude towards proofs, okay? This is important to mention. So, I will put two columns. Column left is: I find proofs — so proofs, I mean rigorous, long, very careful mathematical proofs with all details. These I mean proofs. So, here I will say: I find proof... um... essential, illuminating. You see with the — with this sound I don't need to run the surface [*referring to the chalk sound*], the survey. ... Um... essential, illuminating... um... useful to clarify... um... the best part of any course. And on this column over here I will put: I find the proof unnecessary, confusing. Any other idea? ... Um... I don't know when I have finished the proof or not, right? Because it looks like it sounds convincing but — mysterious, no clear rules of the game.

Survey: Attitude Towards Proofs

The lecturer contrasts two common perspectives on mathematical proofs:

Positive Perspective	Negative Perspective
Essential	Unnecessary
Illuminating	Confusing
Useful to clarify	Mysterious
The best part of the course	No clear rules of the game

Spoken Protocol [00:13:50 - 00:15:00]

And now, okay, this is extreme, I'm making this on purpose. This is extreme, right? This is extreme. I know that any — most of you would be maybe in the middle. But I would like to know which of you lean more towards this opinion or more towards this opinion, okay? So, let's start with this one. So who is here? Who loves proofs and... [*Students cheer and raise hands*] Okay. And here? [*Students cheer and raise hands*] Okay, it's interesting. So it's roughly — there are a bit more people that love proofs, but — but okay. So there is no correct answer, right? So I would be — well, depending on where — in between.

So why am I asking this? So does it coincide? Is it — does it have some correlation with physicist and mathematician or not? Sometimes? What do you think? It could make sense, not always. There are physicist that... but it could make sense. So I think that... um... about this discussion of proofs, I wanted to take ten minutes today to tell you two things. First, why I believe — many people can have many answers — but why I believe that the proofs are important and what will we test in this course, right? Because it's — it's a bit correlated of course.

The first thing is that in — in algebra many times you have a theorem that is very beautiful, very clean, and you can apply in many situations without modifying it, right? In analysis, we will see that, sometimes the theorem that you learned doesn't exactly apply to the situation you — you need. Then you need to know how to modify a little bit the theorem. And the only way to do that is to know the proof and then modify a bit some step here and some step there.

Insight: Computable Truth and the “Lean” Prover

The lecturer concludes by noting that in the age of AI, proofs are increasingly viewed as *computable*. He references “Lean”, a functional programming language and proof assistant. This shifts the definition of a proof from a “*convincing argument*” or an “*opinion*” to a piece of code that either compiles (is correct) or does not. This modern perspective reinforces the necessity of rigorous detail.

End of Introduction

The lecturer concludes the introductory remarks and prepares to begin the first formal topic of the syllabus: the structure of $\mathbb{R}^n$.